

SHANSHAD ALAM

Data Analyst

+91-9661866100 | Shanshad0999@gmail.com | <https://data-analysis-portfolio.onrender.com> | github.com/shah9661

SUMMARY

Data Analyst specializing in Python, SQL, statistical analysis, and data visualization, with experience converting raw and unstructured data into measurable insights. Skilled in data preparation, exploratory analysis, KPI and metric development, predictive modeling, and interactive dashboarding using Power BI, Pandas, NumPy, Scikit-learn, and SQL. Experienced in developing end-to-end analytics solutions across data pipelines, databases, visualization, and deployment.

SKILLS

Languages & Querying: Python, SQL

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI (DAX), Excel

Machine Learning & Statistics: scikit-learn, Statistics, Hypothesis Testing, A/B Testing

Tools & Practices: Git, GitHub, CI/CD (GitHub Actions), PostgreSQL, Streamlit, FastAPI

PROJECTS

Aadhaar Ecosystem Analytics & Decision Support Platform Python PostgreSQL | Pandas | NumPy | Scikit-learn | Power BI
[GitHub](#)

- Engineered a regional maturity classification framework (Expansion/Stabilization/Maintenance) across 36 States/UTs and 804 districts, by developing enrolment-vs-update, Child Inclusion Index (CII) and Biometric Saturation Index (BSI) analytics.
- Improved data quality and geographic consistency by eliminating 591,454 duplicate records and standardizing 57 raw state labels and 932 district labels into 36 States/UTs and 804 districts, using Pandas/NumPy-based cleaning and regex normalization.
- Detected regional and temporal anomalies across 12 months of Aadhaar activity, by applying IQR/Z-score outlier analysis, OLS trend analysis and rolling time-series techniques to identify reporting gaps, activity spikes and child-inclusion imbalances.
- Delivered an interactive Power BI decision-support dashboard for state-level maturity, inclusion and maintenance analysis, by implementing AIMS-based analytics, DAX measures and visualizations from a reproducible Python/Jupyter workflow

Influencer Insights — YouTube Comment Sentiment & Audience Analytics

Python | NLP | Scikit-learn | LightGBM | Visualization | MLflow | DVC | FastAPI | AWS

[GitHub \(Backend\)](#) | [GitHub \(Frontend\)](#)

- Built an end-to-end NLP analytics pipeline to preprocess and classify YouTube comments, transforming unstructured audience feedback into actionable sentiment insights.
- Analyzed audience sentiment using TF-IDF n-gram features and a multiclass LightGBM model, achieving 86.67% weighted F1 and 85.78% macro F1 after class balancing and hyperparameter tuning.
- Identified sentiment patterns across audience comments to help surface positive, negative, and neutral feedback and support content-performance analysis.
- Developed a FastAPI prediction service and Chrome Extension to provide real-time sentiment classification and audience-insight visualization from YouTube comments.
- Dockerized the application and configured GitHub Actions CI/CD for deployment on AWS, creating a reproducible production workflow.

Flight Analytics Dashboard | Python, SQL, Streamlit, Plotly, PostgreSQL

[GitHub](#)

- Built an interactive Streamlit tool for instant city-to-city flight search, replacing manual spreadsheet lookups, backed by a PostgreSQL database with connection-pooled queries (psycopg2).
- Surfaced airline market share, busiest airports, and daily demand trends across the full IndiGo booking dataset through three Plotly visualizations built directly on live query results.
- Designed a fail-safe query layer that returns empty results instead of crashing on bad or missing data, keeping the app stable for end users at all times.
- Deployed a live public demo on Streamlit Community Cloud, making the tool accessible without any local setup.

EDUCATION

B.Tech, Electrical Engineering | Jamia Millia Islamia, New Delhi | CGPA: 7.5

Expected 2027